

Sparks in the det6/det7 run: do they correlate with each other, and do they cross-talk into M3?

(mx17_det6_det7_overnight_6-26-26 / long_run)

MX17 cosmic-bench QA

July 5, 2026

1 Setup

This run reads out *two* detectors under test through the shared DAQ: det6 (FEU3/4) and det7 (FEU6/8), plus the M3 reference telescope (FEU1). A spark = that detector fires > 50 strips. We ask two things: (1) do det6 and det7 sparks correlate event by event, and (2) does a spark in *either* detector inject excess noise into the shared M3 readout (the cross-talk confirmed for det3, companion note `det3_spark_analysis` Q2c)? For (2) we decoded the raw M3 `.fdf` (pulled from `lxplus, DataReader ...analyse`) and joined on `evn` (100% match in a 43 136-event window).

2 det6 and det7 sparks are moderately correlated — a shared muon, not a glitch

Over 71 705 events, det6 sparks 24.0% and det7 31.0% of the time. They are positively correlated but far from lock-step (Fig. 1): So when one detector sparks the other is $\sim 2.4\text{--}2.9\times$ more likely to

	det7 spark	det7 quiet		
			$P(\text{det7} \text{det6 spark})$	56.2%
			$P(\text{det7} \text{det6 quiet})$	23.1% (2.4 $\times$)
			$P(\text{det6} \text{det7 spark})$	43.5%
			$P(\text{det6} \text{det7 quiet})$	15.2% (2.9 $\times$)
det6 spark	9 669	7 536	odds ratio / ϕ	4.3 / 0.31
det6 quiet	12 578	41 922	coincidences vs chance	9669/5338 = 1.8 $\times$

spark, and double-sparks occur 1.8 $\times$ above chance — a real but **moderate** correlation ($\phi = 0.31$). It is *not* a global HV/DAQ glitch: that would force near-total coincidence ($\phi \rightarrow 1$), whereas here most sparks are single-detector (7 536 det6-only, 12 578 det7-only vs 9 669 both). The moderate excess is what a **shared muon** produces: det6 and det7 are stacked in the same telescope, a cosmic crosses both, and each sparks muon-induced (det7 shows a 5.9 $\times$ muon enhancement, companion note) — so their sparks partially coincide through the common muon, without either driving the other. (A small inter-detector DAQ-coupling contribution cannot be excluded, but Sec. 3 shows the shared-readout coupling is only a few-cluster noise effect — far too weak to fake a 50-strip spark.)

3 M3 cross-talk confirmed — with a clean dose-response

Accounting for *both* detectors (“our spark” = det6 or det7 fires), the M3/FEU1 result reproduces det3 and adds a decisive causal handle (Fig. 2):

- **M3 is not knocked out:** it never fires its own spark (`MGv2.Spark`= 0 in all 51 916 decoded events) and drops *no* triggers (100% present).

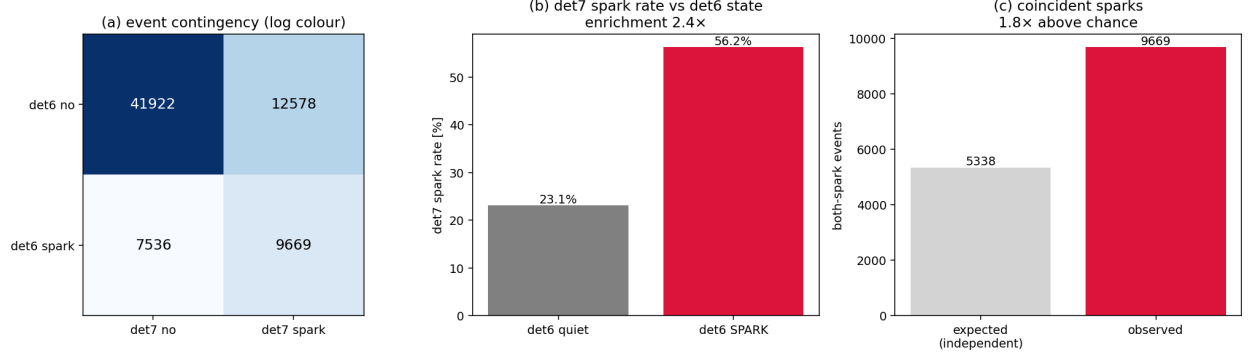


Figure 1: **det6 ↔ det7 spark correlation**. (a) event contingency (log colour); (b) det7 spark rate is 2.4× higher when det6 sparks; (c) coincident double-sparks occur 1.8× above the independent expectation.

- **Excess M3 clusters when we spark:** 11.1 vs 8.8 per event (+26 %) for either-detector sparks.
- **Dose-response:** neither 8.8 → one detector ∼ 10.1 → **both** 12.9 (+47%). The more the shared DAQ discharges, the more spurious M3 clusters appear — a monotonic dose-response is exactly what a cross-talk (noise-injection) mechanism predicts, and hard to explain any other way.

This confirms, on an independent run and with two spark sources, the det3 picture: sparks do not make M3 discharge or lose triggers, but they inject low-level noise into the shared readout in proportion to how much is sparking. That noise adds spurious hits that spoil the M3 straight-line fit, explaining the elevated “no clean M3 track” rate in spark events.

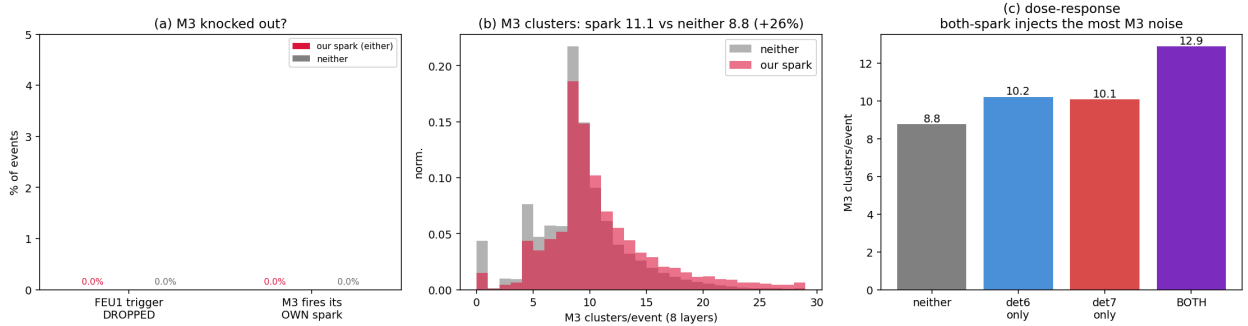


Figure 2: **M3 cross-talk (both detectors)**. (a) M3 never drops a trigger nor sparks (both 0 %); (b) +26 % M3 clusters when either detector sparks; (c) **dose-response** — neither → one → both injects progressively more M3 noise.

4 Takeaways

- det6 and det7 sparks correlate **moderately** (2.4–2.9× conditional, 1.8× coincidence, $\phi = 0.31$) — consistent with a **shared muon** through the stacked detectors, not a global HV/DAQ glitch (which would be near-total) and not a strong inter-detector discharge coupling.
- The **M3 cross-talk is confirmed** on this run counting both detectors’ sparks, with a clean **dose-response** (more sparking → more M3 noise) that nails the causal direction. M3 itself

never discharges or drops triggers.

- Consistent with det3: the cross-talk is a mild, shared-DAQ noise injection — a grounding/shielding issue, not a physical M3 discharge.

Reproduce: `correlate_sparks.py` (local, from `combined_hits`) $\rightarrow$ `d67_events.npz`; FEU1 decoded with `cosmic.bench_m3_tracking`
`DataReader config_d67_cos.json analyse`; then `crosstalk_feu1_d67.py`. Numbers in `correlation.json`, `crosstalk_d67.json`.